AP Chemistry — Trace Packet

Instructions: For each item below, copy the lines under "Copy this exactly" onto a fresh sheet of paper, by hand, in your own natural handwriting. Do not solve the problem yourself or change anything. One item per page. Photograph each page (well-lit, full page in frame, no name on the page). Save each photo as the Item ID shown (e.g. `CHEM-E1.jpg`).

CHEM-E1 (easy)

Problem (for context only — do not solve independently): Write the equilibrium constant expression K for the reaction: 2 NO(g) + O2(g) <=> 2 NO2(g).

Copy this exactly, by hand:

```
K = [NO2]^2 / ([NO]^2 [O2])
```

CHEM-E2 (easy)

Problem (for context only — do not solve independently): A solution has [H+] = 2.5 x 10^-4 M. Calculate the pH.

Copy this exactly, by hand:

```
pH = -log(2.5 x 10^-4) = 3.60
```

CHEM-E3 (easy)

Problem (for context only — do not solve independently): For the reaction C3H8 + 5 O2 -> 3 CO2 + 4 H2O, how many moles of O2 are needed to react completely with 4.0 mol C3H8?

Copy this exactly, by hand:

```
moles O2 = 4.0 mol C3H8 x (5 mol O2 / 1 mol C3H8) = 20.0 mol O2
```

CHEM-H1 (hard)

Problem (for context only — do not solve independently): For $A(g) \rightleftharpoons B(g) + C(g)$, $K_c = 4.0 \times 10^{-3}$ and initial $[A]_0 = 0.50$ M (no B or C present initially). Set up the ICE table and solve for the equilibrium concentration of B.

Copy this exactly, by hand:

ICE: $[A] = 0.50 - x$, $[B] = x$, $[C] = x$

$$K_c = x^2 / (0.50 - x) = 4.0 \times 10^{-3}$$

$$x^2 + 4.0 \times 10^{-3} x - 2.0 \times 10^{-3} = 0$$

$$x = [-4.0 \times 10^{-3} + \sqrt{(4.0 \times 10^{-3})^2 + 4(2.0 \times 10^{-3})}] / 2$$

$x = 0.0428$ M, so $[B] = [C] = 0.0428$ M and $[A] = 0.457$ M

CHEM-H2 (hard)

Problem (for context only — do not solve independently): Given: (1) $C(s) + O_2(g) \rightarrow CO_2(g)$, dH1 = -393.5 kJ; (2) $H_2(g) + 1/2\ O_2(g) \rightarrow H_2O(l)$, dH2 = -285.8 kJ; (3) $C_2H_2(g) + 5/2\ O_2(g) \rightarrow 2\ CO_2(g) + H_2O(l)$, dH3 = -1299.6 kJ. Use Hess's law to find dH for $2\ C(s) + H_2(g) \rightarrow C_2H_2(g)$.

Copy this exactly, by hand:

```
target = 2 x (1) + (2) - (3)

dH = 2(-393.5) + (-285.8) - (-1299.6)

dH = -787.0 - 285.8 + 1299.6

dH = 226.8 kJ
```

CHEM-H3 (hard)

Problem (for context only — do not solve independently): Using the Arrhenius equation k = A e^(-Ea/(RT)), find k given A = 5.0 x 10^12 s^-1, Ea = 75.0 kJ/mol, R = 8.314 J/(mol K), T = 298 K.

Copy this exactly, by hand:

```
k = A e^(-Ea/(RT))

Ea/(RT) = 75000 / (8.314 x 298) = 30.27

k = 5.0 x 10^12 x e^(-30.27)

k = 5.0 x 10^12 x 7.1 x 10^-14

k = 0.36 s^-1
```

CHEM-H4 (hard)

Problem (for context only — do not solve independently): For a first-order reaction, ln[A]t = -kt + ln[A]0. Given k = 0.0250 min^-1 and [A]0 = 0.800 M, find the time t at which [A]t = 0.100 M.

Copy this exactly, by hand:

```
ln[A]t = -kt + ln[A]0

ln(0.100) = -0.0250 t + ln(0.800)

-2.303 = -0.0250 t + (-0.223)

-2.303 + 0.223 = -0.0250 t, so -2.080 = -0.0250 t

t = 83.2 min
```

CHEM-H5 (hard)

Problem (for context only — do not solve independently): A buffer contains 0.30 M acetate (A-) and 0.20 M acetic acid (HA), Ka = 1.8 x 10^-5. Use the Henderson-Hasselbalch equation to find the pH.

Copy this exactly, by hand:

```
pKa = -log(1.8 x 10^-5) = 4.745

pH = pKa + log([A-]/[HA])

pH = 4.745 + log(0.30/0.20)

pH = 4.745 + 0.176

pH = 4.92
```